

The benefit to rebalancing is investor specific. Moving the payoffs from the extreme stock positions back to the center (as in Figure 4.9, Panel D) is optimal for the investor because it cuts back on risk. In our example, the 60% equity/40% bond portfolio turns out to be optimal for an investor with a $\gamma = 0.51$ degree of risk aversion. A certainty equivalent calculation (see chapter 2) reveals that he needs to be compensated 0.29 cents for each dollar of initial wealth for being forced to do the buy-and-hold strategy instead of optimally rebalancing.³⁰ The long-term investor values rebalancing because it reduces her risk and increases her utility. The market does not because there must be other investors who are not rebalancing to take the other side.

Because rebalancing is short volatility, it automatically earns the volatility risk premium. In our example, volatility is constant (the stock volatility is equal to 0.75), but in reality volatility varies over time. Volatility is a risk factor and earns a *negative risk premium*. An investor collects the volatility risk premium by selling options or by being short volatility. I discuss this further in chapter 7.

Viewing rebalancing as a short volatility strategy in moving the payoffs to the center, increasing the losses during extreme low markets, and underperforming the buy-and-hold strategy during extreme high markets makes clear that rebalancing profits from reversals. This is one reason that rebalancing performed well during 1926–1940 and 1990–2011 in Figure 4.5 and 4.6, respectively. When there are strong reversals from the steepest crashes, rebalancing works wonders. We experienced strong reversals after the Great Depression and more recently after the Great Recession and financial crisis in 2008–2009. Conversely, if reversals do not occur, such as in permanent bull or permanent bear markets, then rebalancing will underperform the buy-and-hold strategy.

The fact that rebalancing is equivalent to a short volatility strategy is equivalent to the statement, in the words of Antti Ilmanen, my fellow advisor to the Norwegian sovereign wealth fund, “rebalancing is short regime changes.” Take the extreme case where a regime change occurs and permanently kills equity markets: rebalancing performs poorly because it adds equities as prices decline, and then equity prices are permanently lower. The opposite extreme case is a regime change

and Mark Rubinstein in 1976, which sell stocks as prices decrease—the opposite of rebalancing (see Rubinstein and Leland (1981)). Kimball et al. (2011) develop a model of equilibrium rebalancing. Chien, Cole, and Lustig’s (2012) equilibrium model also has some investors who rebalance and others who do not.

³⁰ The optimal utility for rebalancing is $\frac{1}{1-\gamma} \left[(0.5)^2 \times (2.6896)^{1-\gamma} + 2 \times (0.5)^2 \times (1.2136)^{1-\gamma} + (0.5)^2 \times (0.5476)^{1-\gamma} \right] = 2.3303$ and the optimal utility for buy and hold is $\frac{1}{1-\gamma} \left[(0.5)^2 \times (2.8840)^{1-\gamma} + 2 \times (0.5)^2 \times (1.0840)^{1-\gamma} + (0.5)^2 \times (0.6340)^{1-\gamma} \right] = 2.3270$. The certainty equivalent compensation required by the investor to do buy-and-hold investing instead of optimal rebalancing is $\left(\frac{2.3303}{2.3270} \right)^{\frac{1}{1-\gamma}} - 1 = 0.29$ cents per dollar of initial wealth. Notice that these are the only calculations where we actually use the real-world 0.5 probability of an upward move in the tree. All the option valuations are done using risk-neutral probabilities.